Reforming the UK Triple Lock: The State Pension under Forecast Uncertainty*

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Abstract

Uprating rules that take the maximum of several indices, such as the UK State Pension triple lock, carry higher rises forward, so their cost depends on the volatility of prices and earnings, not only on their expected path. We estimate the fiscal and distributional effects of replacing the triple lock from 2030 with a rule that keeps its 2.5% floor but tracks earnings over time, as the UK government announced in September 2026. We combine stochastic paths of the statutory inputs, calibrated to official forecasts and screened by a test registered in advance, with full microsimulations in PolicyEngine UK. The provisional expected saving in 2039-40 is £11.1bn gross and £7.9bn net, against £0.9bn on the central forecast. Costing indexation reforms on a single forecast path can understate their fiscal effect.

Keywords: pension indexation, triple lock, microsimulation, forecast uncertainty, stochastic simulation, public finances
JEL codes: H55, H68, C63, E37

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1 Introduction

The triple lock has raised the basic State Pension each April since 2011, and the new State Pension since it began in 2016, by the highest of CPI inflation, earnings growth and 2.5%. Because it takes the highest of the three each year, a year in which prices or the floor run ahead of earnings raises the pension for good, and later rises build on the higher level. The Office for Budget Responsibility (OBR) assumes in its long-term projections that the triple lock raises the pension by 0.6 percentage points a year more than earnings (Office for Budget Responsibility, 2026d).

The Prime Minister set out a change from April 2030 in his speech to Labour's conference on 29 September 2026. As the BBC reported it, he said he would "adjust" the triple lock, that the pension would rise every year at least by inflation or 2.5%, and that "it will hold its value relative to earnings over time" (BBC News, 2026). DWP's State Pension uprating analysis, published the same day, says the pension will reach "a record high relative to earnings over the coming three years" and from April 2030 will retain "that record high value relative to earnings": "In any given year it will go up by at least inflation or 2.5% – and anything more that is needed to retain that value" (Department for Work and Pensions, 2026b). DWP does not state the measures of inflation and earnings or the base year. We take the triple lock's own measures, September CPI and May–July average weekly earnings, and the 2029-30 level as the base, and call the result the Burnham plan.

The paper follows the order of the dashboard. Section 4.3 gives what the plan saves across the model's paths, and how that compares with DWP's and other published costings. Section 2 shows how the triple lock works and what the plan would have paid had it started earlier. Section 4.1 and Section 4.2 run both rules on the OBR's central forecast and on simulated paths. Section 4.4 and Section 4.5 follow the whole population and example pensioners on one path. Section 3 describes the method, the inputs, our assumptions and the limitations.

Every fiscal and household figure is a full run of PolicyEngine UK (PolicyEngine, 2026a,b). We use the release bundle policyengine 6.2.6, which pins policyengine-uk 2.125.1 and policyengine-core 3.33.0 and is published for the Enhanced Family Resources Survey (FRS) for 2024-25, release 1.58.0 (52,846 household records). The survey population is reweighted to each year (Section 3.3). A path's CPI and earnings growth enter the model's economic assumptions before the data load, so benefit rates, CPI-indexed tax thresholds and earnings follow it. The State Pension flat rates are then set from each rule. Weekly amounts, rates and the spread of the pension gap across paths are rule arithmetic and are labelled as such. All money is in cash terms. Fiscal years are named by their first year in figures ("2039" is 2039-40).

2 The triple lock and the Burnham plan

2.1 The rule and its inputs

Each April the full basic and new State Pension rise by the highest of three figures: the CPI 12-month rate for the previous September (Office for National Statistics, 2026c); growth in average weekly earnings (AWE), whole-economy total pay, in May to July of the previous year over the same months a year earlier (Office for National Statistics, 2026b); and 2.5%. The law requires the basic and new State Pension to rise at least in line with earnings (Social Security Administration Act, 1992); the prices leg and the floor are policy on top. ONS publishes both inputs to 0.1 point, and the rules here take them at that precision, so the triple lock's rate is

exactly the larger input or the floor. Only the flat-rate pensions are triple-locked. The additional State Pension rises with September CPI under both rules here, as DWP's analysis assumes for the parts of the pre-2016 pension that the triple lock does not cover (Department for Work and Pensions, 2026b).

2.2 The triple lock since 2011

Figure 1 and Table 1 replay the rule on the published inputs for each April from 2011 to 2026, using the latest ONS figures. CPI set the rise six times, earnings six times and the floor four times. Over those 16 rises the triple lock raised the pension by 88% in all, against 66% for earnings alone, 60% for prices alone and 48% for 2.5% a year. Carrying those higher rises forward leaves the pension 12.7% above where an earnings link alone would have taken it.

In April 2022 the law suspended the earnings leg, whose May–July input was 8.4%, so that April's rise was CPI (3.1%) (Social Security (Up-rating of Benefits) Act, 2021); the replay does the same, though the earnings-alone line uses the published figure. The replay differs from the rise actually paid in 6 of 16 years (April 2011: paid 4.6%, rule 3.1%; April 2016: paid 2.9%, rule 2.8%; April 2019: paid 2.6%, rule 2.7%; April 2020: paid 3.9%, rule 4.0%; April 2025: paid 4.1%, rule 4.4%; and April 2026: paid 4.8%, rule 4.9%), because the rises paid used the figures first published, and April 2011's rise, set by that year's up-rating order, was larger than the rule gives (Social Security Benefits Up-rating Order, 2011).

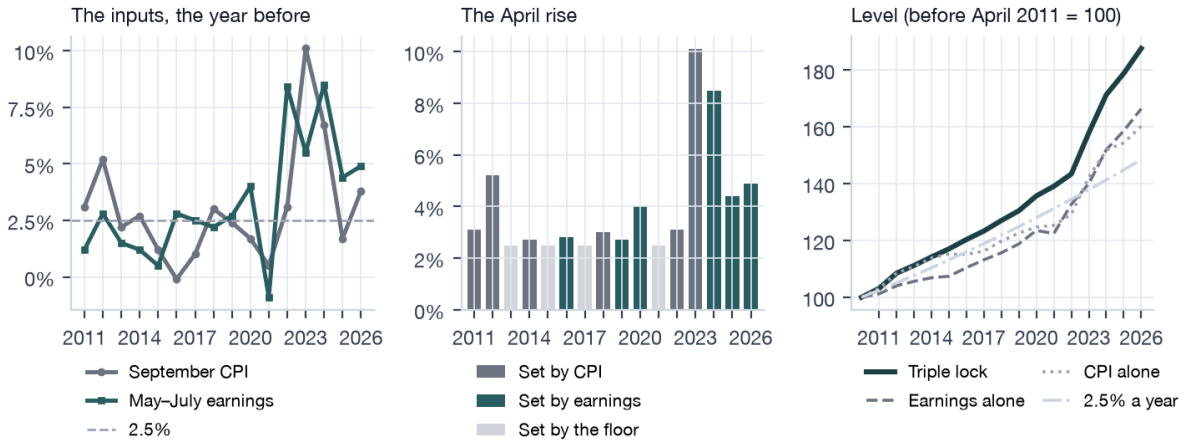


Figure 1: The triple lock, April 2011 to April 2026, on the latest ONS figures. Left: September CPI and May–July earnings growth in the year before each April (for April 2022, the published earnings growth), and the floor. Centre: the rise, coloured by the input that set it (April 2022's earnings leg suspended, as in law). Right: the cumulative level under the triple lock, earnings alone (published figures), CPI alone and the floor alone.

2.3 The Burnham plan's rule

The Burnham plan follows the triple lock to April 2029. DWP's description is that from then on "In any given year it will go up by at least inflation or 2.5% – and anything more that is needed to retain that value", where the value is the pension's record high relative to earnings (Department for Work and Pensions, 2026b). We write this as follows. From April 2030 let L_t

Table 1: The triple lock replayed on the published inputs. Inputs are for the year before each April; 'rise paid' is the rise in the basic State Pension actually paid.

April	September CPI	May–July earnings	Triple lock rise	Set by	Rise paid
2011	3.1%	1.2%	3.1%	CPI	4.6%
2012	5.2%	2.8%	5.2%	CPI	5.2%
2013	2.2%	1.5%	2.5%	the 2.5% floor	2.5%
2014	2.7%	1.2%	2.7%	CPI	2.7%
2015	1.2%	0.5%	2.5%	the 2.5% floor	2.5%
2016	−0.1%	2.8%	2.8%	earnings	2.9%
2017	1.0%	2.5%	2.5%	the 2.5% floor	2.5%
2018	3.0%	2.2%	3.0%	CPI	3.0%
2019	2.4%	2.7%	2.7%	earnings	2.6%
2020	1.7%	4.0%	4.0%	earnings	3.9%
2021	0.5%	−0.9%	2.5%	the 2.5% floor	2.5%
2022	3.1%	8.4%	3.1%	CPI	3.1%
2023	10.1%	5.5%	10.1%	CPI	10.1%
2024	6.7%	8.5%	8.5%	earnings	8.5%
2025	1.7%	4.4%	4.4%	earnings	4.1%
2026	3.8%	4.9%	4.9%	earnings	4.8%

be the pension's level and A_t an earnings path that starts at the 2029-30 level and grows with May–July earnings, $A_t = A_{t-1}(1 + e_{t-1})$. Then

$$L_t = \max\big(L_{t-1}\,(1 + \max(c_{t-1}, f)),\ A_t\big),$$

where c is September CPI, e is May–July earnings growth and f is the 2.5% floor. The pension never rises by less than the higher of CPI and 2.5%, and never falls below its earnings path. The top-up to the earnings path is rounded up to 0.1 point, so rounding never leaves the pension below it. Property tests in the repository check, for all inputs, that the plan meets both guarantees, that its rate is the smallest that does, and that before rounding its level never exceeds the triple lock's.

Two features follow from the rule. In its first year, April 2030, the plan always pays what the triple lock pays: its earnings path starts from the 2029-30 level, so the top-up is exactly the earnings rise whenever earnings lead. After that the plan pays less only after some earlier year in which the higher of CPI and 2.5% ran ahead of earnings. Both rules pay that higher figure then, but it lifts the plan above its earnings path; when earnings next lead, the plan rises only by the higher of CPI and 2.5% until earnings catch up, while the triple lock rises with earnings from the higher level.

3 Data and method

3.1 A path through PolicyEngine UK

A path is calendar-year CPI and earnings growth for 2027–2039, plus the statutory inputs that set the April rises. The calendar growth replaces the model's OBR growth assumptions for CPI and average earnings, with RPI and CPIH moved by the same amount as CPI. The statutory inputs, September CPI and May–July earnings growth, replace the model's own, from which

policyengine-uk builds its triple lock. Both enter in a scenario applied before the data load. The model builds its derived series (uprating indices, the lagged CPI that uprates benefits, its own triple lock) when it loads parameters, so the same changes passed as a reform after loading would do nothing. policyengine-uk's derived series now run to 2039-40 and beyond by themselves. The analysis extends only the uprating of private pensions, which policyengine-uk still stops at 2034.

Each run checks, in every year, that the model follows the path, and fails if any check misses. Benefit rates uprated by CPI must follow the path's calendar-year CPI of the year before. In law September CPI uprates them, so this is a simplification. CPI-indexed thresholds must follow the path in the same year, and employment income must follow the path's earnings. The model's own statutory inputs must be the path's, and its own triple lock must be the larger of them and the floor. Every flat-rate pension must scale exactly by the ratio of the two rules' amounts. The model's adjustment of pay for changes in employer National Insurance must be zero. The Pension Credit guarantee must follow the path's earnings. The variables the net saving is split into must add up to the model's own government balance. A separate identity must hold too: the government's balance plus households' net income must equal market income less pension contributions, after adjusting for items counted on only one side. Private pension income moves with the previous year's RPI up to a cap. Dividend, property, savings and self-employment income, rents and council tax do not follow the path, and each run records their growth. April 2027's benefit uprating is the model's own forecast of calendar-2026 CPI on every path, not September 2026 CPI.

3.2 The model and data release

Every run uses one release bundle: policyengine 6.2.6, which pins policyengine-uk 2.125.1 and policyengine-core 3.33.0, with the Enhanced FRS for 2024-25, release 1.58.0. policyengine.py publishes the bundle and states that this data release works with this model release. The data release carries the Pension Credit and pension-age Housing Benefit fixes made to the data after the earlier pilot. Each run records the installed versions and the dataset's hash. The build refuses to start any model job unless the installed packages, the dataset's bytes and the release manifest all match the bundle. An earlier pilot of this method ran on policyengine-uk 2.120.0 with an older data release. That pair was never released together, so we quote none of its figures.

One change in the bundle matters directly for the net saving. Housing Benefit treats a person who receives the Pension Credit guarantee credit as having no income or capital to count, a passport to the maximum award. The law keys this on receipt of the guarantee credit. Until policyengine-uk 2.123.6 the model keyed it on entitlement, so a pensioner entitled to the guarantee credit but not claiming it still got the passport (policyengine-uk #1927). In our runs a lower pension makes some households newly entitled to the guarantee credit. Keyed on entitlement, they moved straight to the maximum Housing Benefit, which overstated the Housing Benefit offset. The bundle keys all three parts of the passport (applicable income, tariff income and capital) on receipt. Before any model job the build tests this on synthetic pensioners. One is entitled to the guarantee credit but does not claim it and must face the ordinary Housing Benefit means test. One claims it and must get the maximum award. The check also reverses the receipt flag, to confirm that every part of the passport reads it. If any check fails, the build stops.

3.3 The population in each year

The survey is a sample of households in 2024-25. We age it statically. Each record stands for a person of its age in every year, not a survivor aged forward. So a record aged seventy-five stands for someone aged seventy-five in 2024-25 and for someone aged seventy-five in 2039-40. Three things change from year to year: the weight each household carries, the age a top-coded record represents, and each person's State Pension type.

Enhanced FRS ages are top-coded at 80. We give each top-coded record a represented age from 80 to 105. Within each sex the records are put in a fixed order by a hash of the data, and their weights are split across ages in proportion to the ONS's single-year population shares. The weighted number of people aged 80 and over does not change. Every record also gets a fixed birthday within its year of age, drawn as policyengine-uk draws it, so records that keep their survey age keep the model's own birthday and pension type. Anything the model derived from the survey ages is cleared, so the model's own State Pension age and every reader of it see the represented ages.

Household weights are the dataset's own up to 2025, the year the data release calibrates its weights to. After that we reweight households so that each sex and age group grows as it does in the ONS's 2024-based principal population projection for the UK (Office for National Statistics, 2026e). The groups are five-year bands up to 59, single years from 60 to 79, then 80 to 84, 85 to 89, and 90 and over. A fiscal year's population is three-quarters of the mid-year population of its first year and a quarter of the next. The targets are the survey's own counts in 2025 grown at the ONS rates, not the ONS levels. The reweighting finds the weights closest to the 2025 ones, in relative entropy, that hit every target. No weight may fall below a fifth of its 2025 value or rise above five times it. If a target cannot be reached within those bounds, the run fails. Person and benefit-unit weights follow their household's.

Each person's State Pension type follows from their date of birth, from the represented age and birthday in each year. Men born before 6 April 1951 and women born before 6 April 1953 reached State Pension age before 6 April 2016 and are on the basic State Pension. People born on or after those dates are on the new State Pension, and people below State Pension age have none. So in later years more pensioners are on the new State Pension. A test checks that these types equal policyengine-uk's own on the same ages and birthdays. State Pension age is the model's own, set from each person's date of birth by the law's timetable, including the rise to 67 for people born on or after 6 April 1960 (Pensions Act, 2014). Because ages are held, a record's date of birth moves a year later each year. People aged 66 are partly over State Pension age in 2026-27 and 2027-28 and below it from 2028-29.

The State Pension a record reports in the survey has to be split between these types each year. The reported amount counts only for people over State Pension age in the survey year, since nobody is paid a State Pension before it. A few records below that age report one. They are reporting errors, and the model already pays them nothing. In each year the counted amount is split by that year's type at the type's survey-year flat rate. The flat-rate part moves with each rule. The rest is the additional State Pension (and, on the new State Pension, protected payments). We hold it at its survey-year amount, grown by September CPI. Splitting by each year's type matters. Split by the survey year's type, a record moved from the basic to the new State Pension would be paid the difference between the two flat rates twice, once inside its new State Pension and again as additional pension. Every run checks that, in the survey year, basic, new and additional State Pension add up to the counted report for every person, to £0.01.

A record that reported a basic State Pension and is a new State Pension recipient in a later

year raises a question: what amount should it get? By default it keeps its reported amount, split at the new type's flat rate. As a bound, a sensitivity raises only that record's flat-rate part to the full new State Pension under each rule. We cannot do better from the survey. In law the new State Pension of someone with pre-2016 contributions starts from a foundation amount: the higher of the old-system and new-system calculations, less any contracting-out deduction, plus later qualifying years up to the full rate (Pensions Act, 2014). That needs each person's qualifying years before and after 2016, their additional pension history and their contracting-out record. The Enhanced FRS has none of these. It has only the total State Pension each person reports. So the law does not fix an amount for a re-typed record here. The full-rate bound limits the flat-rate part only. It is not an estimate of the transitional entitlement, and not a confidence bound. Every result in this release keeps the reported amount; the full-rate bound has not yet been rerun on this release bundle and will be reported with the post-Budget release.

We measure what each piece of the ageing does with paired full runs on the same paths. One treatment holds the survey weights and types with represented ages. One adds the ONS reweighting, one adds the cohort types, and one adds both, which is what every result in this paper uses. The difference between the last and the sum of the middle two is their interaction. A further control reweights only to the ONS total population, not by age and sex. The four-way design has not yet been rerun on this release bundle; its results will follow with the post-Budget release.

3.4 How the monthly model makes the paths

The paths come from a model of the months themselves, since the triple lock's inputs are measured over different months from the calendar-year figures the OBR forecasts. It is a vector autoregression on the monthly log changes in the CPI index (Office for National Statistics, 2026d) and in the whole-economy AWE total pay level (Office for National Statistics, 2026a), with an intercept and month dummies, fitted by least squares from 2000 to the latest published month. The lag order is chosen by the Bayesian information criterion (it picks one). The furlough months of 2020–21 are left out of the fit and of the pool of shocks, because their earnings changes reflect who was paid rather than what pay did. Each simulated month draws a pair of fitted residuals at random, so the shocks to prices and earnings keep their joint distribution; the one month for which CPI is published but AWE is not draws its earnings shock from its normal distribution given the observed CPI shock. From the simulated months come both the statutory inputs (September CPI, May–July earnings growth) and the calendar-year growth that the rest of the model uses, so the two stay consistent on every path. Calendar-year AWE total pay stands in for the OBR's measure of average earnings.

The model's own averages sit well away from the OBR's forecast, so every path is shifted by the same monthly drift: the smoothest drift that makes the paths' mean calendar-year CPI and earnings growth equal the central path in every year from 2027 to 2039. It starts in September 2026, the first month for which neither series is published. The drift is not small: it moves earnings growth by −1.51 points a year in 2027 and CPI growth by −0.60 points a year by 2039. Because every path moves by the same amount, each path's ups and downs, and the link between prices and earnings, remain the model's own. Tilting the weights of the paths to the same means instead (Robertson et al., 2005) would leave far fewer effective paths, and a drift held constant within each year would oscillate between years and put false reversals into the statutory inputs.

The same model with Student-t or Gaussian shocks, the same model with two lags, and an

annual model are scored beside it in the screen of Section 3.6. They are not run in full. The screen keeps this model as the primary if it passes, and never promotes another form in its place.

3.5 Choosing the calibration

The paths could be calibrated in more than one way. Besides matching the OBR's means, they could also be tilted, by entropy tilting (Robertson et al., 2005), so that the variance of the gap between May–July earnings and September CPI, and how often the lead passes between them, match history. We chose between these on two tests of the expected gap between the two rules, not of the saving, since the gap is what the calibration changes.

The first is a chronological backtest over 12 OBR forecasts in the OBR's historical official forecasts database (Office for Budget Responsibility, 2026c), from the June 2010 Budget forecast to the March 2021 EFO. At each, the monthly model is fitted only on data to the December before the forecast, and the calibrations take the forecast's calendar-year CPI and earnings as the means; the plan starts at the April rise two years after the forecast, and the gap is measured after four rises. This backtest uses today's published data and is not rerun on the Budget-day inputs. Table 2 compares each method's expected gap with what happened, scoring April 2022 both as in law (earnings suspended) and with the published earnings. Of the calibrations that match the OBR's means, which the expected value needs, the drift shift and entropy tilting to the means have almost the same bias: $+0.20 \pm 0.45$ and $+0.21 \pm 0.45$ points of the pension as in law, each within one standard error of zero, and -0.60 ± 0.57 and -0.60 ± 0.57 as published, about one standard error. We use the shift because tilting keeps so few effective paths. Tilting further, to past dynamics, made the bias larger under both treatments (-0.35 ± 0.41 and -1.15 ± 0.52), though its mean absolute error as in law is lower; we chose on bias because the expected value is a mean. The uncalibrated model has the smallest bias as in law (-0.01 ± 0.41) but does not match the OBR's forecast. The OBR forecast used as a single path has the largest bias under both (-0.86 ± 0.41 and -1.67 ± 0.54): it expects too few changes of lead. The origins overlap, sharing three of four target years with their neighbours, so the standard errors, computed as if they were independent, understate the uncertainty; the data are today's revised figures, and the model design was chosen with the whole sample in view.

The second, the past-years check of Section 2, cannot test the drift shift, since no forecast covers those years; it bears only on tilting to past dynamics, which put what happened at the 93rd percentile of its paths, against the 56th for the model alone.

The tilts are kept as sensitivities, computed by reweighting the same full runs (Table 3). Across the 12 of them the expected gross saving in 2039-40 runs from £5.59bn to £12.41bn, against £11.14bn for the primary calibration. Reweighting leaves few effective runs in some cases, as few as 43. Those rows' standard errors are approximate and too narrow. A reweighting with fewer than 100 effective runs keeps its row but is left out of any range we quote. The tilts did worse in the backtest, so this rule promotes none of them.

Realised gap, averaged over the forecasts: 0.98% as in law, 1.79% with April 2022 as published.

3.6 The screen set before scoring

An expected value needs a model of prices and earnings whose paths behave like the statutory inputs. We wrote a screen for this and committed it before scoring any model. We call it C1. It scores five forms, fixed in advance. The primary is the monthly model of Section 4.2, with one

Table 2: The expected-value backtest: mean expected gap between the triple lock and the Burnham plan after the rises, % of the pension, and its bias (± standard error, treating the origins as independent) against what happened, with April 2022 as in law and as published.

Method	Expected gap	Bias, in law	Bias, published	Mean absolute error, in law	Lead changes, predicted minus actual, in law
Not calibrated	0.97	−0.01 ± 0.41	−0.82 ± 0.53	1.05	0.03
Tilt to the OBR means	1.19	+0.21 ± 0.45	−0.60 ± 0.57	1.26	−0.05
Drift shift (used here)	1.19	+0.20 ± 0.45	−0.60 ± 0.57	1.26	−0.06
Drift shift, then tilt to past dynamics	0.63	−0.35 ± 0.41	−1.15 ± 0.52	0.95	−0.82
OBR forecast as one path	0.12	−0.86 ± 0.41	−1.67 ± 0.54	1.01	−1.08

Table 3: The expected saving in the final year under other calibrations, £ billion (± 1.96 SE), from the same full runs reweighted. Each row keeps the OBR's means and tilts the shifted paths to a history window's gap variance and rate of lead changes (and, where shown, its share of years below the floor), with the furlough years and April 2022 treated as the Treatment column says. Effective runs measure how many of the draws the reweighting leaves in use.

History	Treatment	Tilted to	Gross	Net	Effective paths	Effective runs
Primary: drift shift only			11.14 ± 0.13	7.86 ± 0.09	50,000	160
2001-2025	2020-21 left out	variance, leads	7.05 ± 1.80	4.98 ± 1.28	9,869	61
2001-2025	2020-21 left out	variance, leads, floor share	5.59 ± 1.98	3.95 ± 1.40	3,526	43
2001-2025	2022 as in law	variance, leads	7.14 ± 1.73	5.04 ± 1.23	11,717	67
2001-2025	2022 as in law	variance, leads, floor share	6.32 ± 1.91	4.47 ± 1.35	6,594	52
2001-2025	2022 as published	variance, leads	9.84 ± 1.17	6.94 ± 0.83	33,503	115
2001-2025	2022 as published	variance, leads, floor share	8.95 ± 1.54	6.32 ± 1.09	19,595	83
2010-2025	2020-21 left out	variance, leads	9.11 ± 1.03	6.43 ± 0.73	33,079	118
2010-2025	2020-21 left out	variance, leads, floor share	8.52 ± 1.38	6.02 ± 0.98	22,714	90
2010-2025	2022 as in law	variance, leads	8.42 ± 1.13	5.95 ± 0.80	29,417	111
2010-2025	2022 as in law	variance, leads, floor share	8.33 ± 1.23	5.89 ± 0.88	27,808	104
2010-2025	2022 as published	variance, leads	12.41 ± 0.95	8.75 ± 0.66	41,969	141
2010-2025	2022 as published	variance, leads, floor share	12.38 ± 1.08	8.73 ± 0.75	40,733	140

lag and bootstrapped shocks. The others are the same model with two lags, the same model with Student-t shocks joined by a t-copula, the same model with Gaussian shocks, and an annual model of calendar CPI and earnings that adds blocks of past gaps between the statutory and calendar measures. Every form is shifted to the same OBR means. Each is fitted only on data from before each forecast origin and scored on what followed.

There are two tests. Test A uses the six spring forecasts from 2016 to 2021 that have two complete earlier blocks of forecast errors. Test B uses all twelve complete spring forecasts from 2010 to 2021. Each test scores four things over the four years after the origin: the gap between earnings and CPI in each year, the number of changes of lead, the share of years with earnings below the floor, and the gap the two rules open after four rises. A form passes only if it meets every condition. Its central 80% bands must cover between half and all of the outcomes, for the yearly gaps and for the final gap, and the band for the final gap must be consistent with 80%. Its biases must be small: at most 1.5 percentage points on the gap, one change of lead per path and a quarter on the floor share. Each of its proper scores must be no more than 1.25 times the primary's, so that a form cannot pass by spreading its paths wider. A version fitted on data before 2011, with no calibration, must put the gap that actually opened over the rises of 2012 to 2026 between the 5th and 95th percentiles of its paths. Every path and score must be finite. C1 required all this under both treatments of April 2022: with the published earnings, and with the earnings leg suspended as the law applied it. The appendix gives the scores (Section A).

No form passed C1. All five failed test A with the published earnings. The band for the final gap covered what happened at only one of six origins. The cause is the furlough base effect: earnings growth in May to July 2021 was measured against pay that furlough had held down a year earlier. For the primary this was the only failure. The other forms had further failures. Under C1 as written, no model was adequate and no expected value was allowed.

We then changed the rule, after seeing these scores, and we say so here. The analysis models the statutory rule, and Parliament suspended the earnings leg for April 2022 (Social Security (Up-rating of Benefits) Act, 2021). The new rule, C2, scores April 2022 as the law applied it. It was written and committed on its own, at commit 65343e2 of the repository, before any code for it was written and before any C2 score. That commit also records the C1 failure. C2 changes two things. First, in the year that set April 2022's rise, earnings are set equal to CPI, in both the simulated paths and the outcomes. Second, any statistic that this legal rule fixes in both is left out of the scores. Cells are chosen by the legal rule, never by their values. In test A this removes four of the 24 yearly gap cells, the 2021 gap in each window that contains it. The final gap keeps the suspended year in its arithmetic. Every C1 threshold stays as it was, now applied to the suspended treatment in both tests. Scores with the published earnings are reported beside every C2 result, but they have no say in the verdict. C2 does not turn the C1 failure into a pass, and both records stay in the repository.

The binding C2 score uses the inputs available after the Budget: September 2026 CPI, May to July 2026 earnings, and the OBR's forecast of 28 October 2026. A score on earlier inputs is a dry run and allows nothing. If the original primary passes the binding score, the release keeps a model-conditional expected value beside the single scenarios. If it fails, the build drops the expected value automatically and keeps the scenarios alone, with the failure recorded. No other form is promoted in its place, and forms are never averaged. In a dry run on today's inputs the original primary passes C2: in test A the outcome fell inside its central 80% band in four of six final years and 80% of the 20 annual checks. This paper reports that provisional result; the binding score follows the 28 October 2026 Budget.

3.7 Strata and estimator

The 50,000 paths are not all run. Paths on which the two rules pay the same rise every year save exactly nothing, since both runs get identical inputs. They are 0.79% of paths and form their own stratum. One extra path from it is run to confirm that it saves nothing. The rest are split into 11 strata, each holding an equal share of the paths, by the gap in the full new State Pension in 2039-40. The rule arithmetic gives that gap for every path before any model run. The design has 160 sample slots. They are allocated across the strata in proportion to each stratum's share of paths times the spread of the gap within it (Neyman, 1934), with at least two in every stratum. Two of them sample the stratum of identical rules, so that its weight stays in the estimate if a mean-path scenario makes its saving nonzero. Slots are drawn at random with replacement within each stratum. A path drawn twice is run once and counted twice. The slots give 160 distinct paths, each a full run (Table 4). The spread of the gap stands in for the spread of the fiscal saving, which is not known before the runs.

For any output y the estimate is $\hat{\mu} = \sum_h W_h \bar{y}_h$, where W_h is stratum h's share of the paths and $\bar{y}_h$ the mean over its slots. Its Monte Carlo variance has two parts, and we report them separately. The first is path sampling, $\sum_h W_h^2 s_h^2 / n_h$, with s_h the standard deviation and n_h the number of slots in the stratum. The second is the first phase, $\sum_h W_h [s_h^2 + (\bar{y}_h - \hat{\mu})^2]/N$, where N is the number of simulated paths. It measures how far the simulated paths themselves could fall from the model's full distribution. The stratum of identical rules counts in the second sum. The standard error we quote is the square root of the two together. Neither part measures the choice of model, which the single scenarios show instead. Tests in the repository check that the estimator is unbiased, that its interval covers about 95% of the time on synthetic cases, and that the published estimates recompute from the per-path records in the results file.

Table 4: The strata: each an equal share of the paths on which the rules differ, by the final-year weekly gap (rule arithmetic), and the draws in each (a path drawn twice is run once).

Stratum	Share of paths	Draws	Distinct paths
0	0.79%	2	2
1	9.92%	20	20
2	9.92%	11	11
3	9.92%	9	9
4	9.92%	8	8
5	9.92%	7	7
6	9.92%	8	8
7	9.92%	9	9
8	9.92%	11	11
9	9.92%	16	16
10	9.92%	59	59

4 Results

4.1 The OBR's central forecast

4.1.1 The central path and its sources

The central path uses the OBR's March 2026 Economic and fiscal outlook (EFO) for calendar-year CPI and average earnings growth in 2027–2030 (Office for Budget Responsibility, 2026a), and its long-term economic determinants for 2031–2039, converted from fiscal to calendar years (Office for Budget Responsibility, 2026d). After 2030 the OBR describes the determinants as a long-term projection, not a forecast. In the determinants calendar-year CPI settles at 2.0% and earnings growth rises to 3.7% by 2039.

The inputs to each April's rise are taken as close to the statutory timing as the forecasts allow (Table 5). April 2027's rise uses the published May–July 2026 earnings growth, 3.9%, and August 2026 CPI, 3.1%, standing in for September's, which ONS publishes on 21 October 2026. CPI would have to rise by more than 0.8 points from August to September to overtake earnings growth; its largest August-to-September rise since 1997 is 0.7 points. The rises of April 2028, 2029, 2030 and 2031 use the EFO's September-quarter CPI and April–June earnings growth, and later rises the calendar-year path, for which no quarterly forecast exists.

4.1.2 The two rules on the central path

Figure 2 shows both rules on this path. Earnings grow by less than 2.5% for several years, so the floor sets the triple lock in April 2028, 2029, 2030 and 2031. From April 2032 earnings lead every year. The two rules differ in two years: April 2032 (triple lock 2.7%, set by earnings; plan 2.5%, set by the 2.5% floor) and April 2033 (triple lock 3.0%, set by earnings; plan 2.8%, rising to its earnings path). By 2039-40 the full new State Pension is £359.85 a week under the triple lock and £358.45 under the plan, 0.4% lower.

4.1.3 Why the saving is small on this path

The whole difference on the central path comes from the floor years just after the switch, April 2030 and 2031. The plan pays the floor then too, but earnings grow by less, so the plan's pension ends up above its earnings path. When earnings pick up, the plan rises by only the floor, and then by what brings it back to its earnings path, while the triple lock rises with earnings from the higher level. Once the plan is back on its earnings path both rules rise with earnings, and the gap stays where the floor years left it. The plan saves £0.66bn in State Pension spending in 2034-35 and £0.86bn in 2039-40; net of income tax and benefits, £0.46bn and £0.62bn (Table 5).

On a single path like this the triple lock rises with earnings in the long run: the carry-forward pays out only when the lead passes between earnings and the higher of CPI and 2.5%. The central path has no such reversals after 2032. The OBR's long-term projections instead assume a triple-lock premium of 0.6 points a year over earnings (Office for Budget Responsibility, 2026d); Section 5.3 compares that with the premium across our paths.

4.2 Individual paths

4.2.1 A random path

The central forecast is one path of many. Figure 3 runs both rules on a path drawn at random from the 50,000 paths behind the expected saving of Section 4.3: draw 8,580 of 50,000, picked at random (seed 20261001); its 2039-40 gap is £1.96 a week, at the 2nd percentile of those paths: 98% of them open a bigger gap by then. We keep it because it was drawn at random. The middle and 90th-percentile paths below give the context.

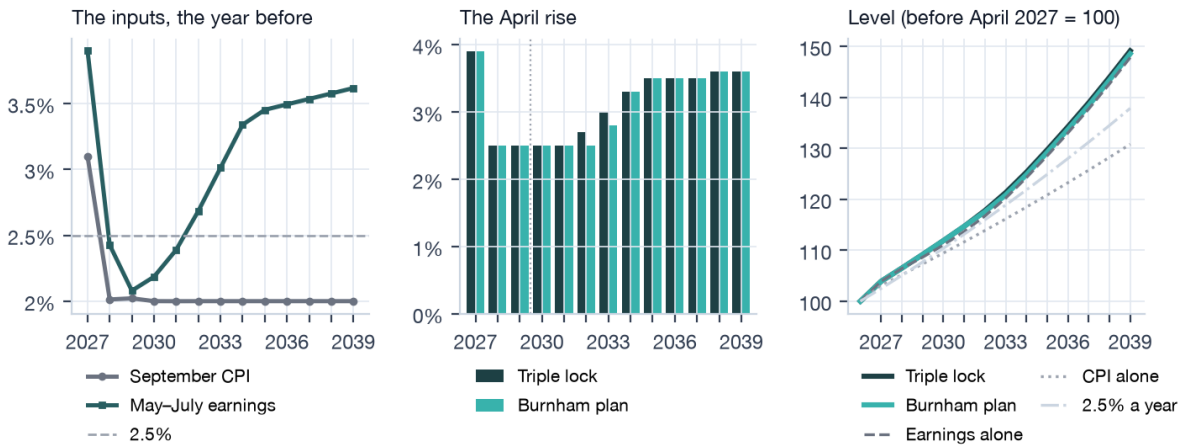


Figure 2: The two rules on the OBR's central forecast. Left: the inputs behind each April's rise (for their sources see the text). Centre: the April rise under each rule; the dotted line marks the switch. Right: the level of the full new State Pension under each rule, and under earnings alone, CPI alone or the floor alone. Rule arithmetic.

Table 5: The central path year by year: the inputs (for the year before each April), each rule's rise and what set it, the full new State Pension a week under each rule (rule arithmetic), and the saving in the fiscal year that starts that April (full model runs on the Enhanced FRS, £ billion).

April	CPI	Earnings	Triple lock	Burnham plan	Weekly, triple lock	Weekly, plan	Gross	Net
2027	3.1%	3.9%	3.9% (earnings)	3.9% (triple lock)	£250.71	£250.71	0.00	0.00
2028	2.0%	2.4%	2.5% (floor)	2.5% (triple lock)	£256.98	£256.98	0.00	0.00
2029	2.0%	2.1%	2.5% (floor)	2.5% (triple lock)	£263.40	£263.40	0.00	0.00
2030	2.0%	2.2%	2.5% (floor)	2.5% (floor)	£269.99	£269.99	0.00	0.00
2031	2.0%	2.4%	2.5% (floor)	2.5% (floor)	£276.74	£276.74	0.00	0.00
2032	2.0%	2.7%	2.7% (earnings)	2.5% (floor)	£284.21	£283.66	0.30	0.20
2033	2.0%	3.0%	3.0% (earnings)	2.8% (earnings path)	£292.74	£291.60	0.63	0.45
2034	2.0%	3.3%	3.3% (earnings)	3.3% (earnings path)	£302.40	£301.22	0.66	0.46
2035	2.0%	3.5%	3.5% (earnings)	3.5% (earnings path)	£312.98	£311.76	0.70	0.50
2036	2.0%	3.5%	3.5% (earnings)	3.5% (earnings path)	£323.93	£322.68	0.74	0.52
2037	2.0%	3.5%	3.5% (earnings)	3.5% (earnings path)	£335.27	£333.97	0.78	0.57
2038	2.0%	3.6%	3.6% (earnings)	3.6% (earnings path)	£347.34	£345.99	0.82	0.60
2039	2.0%	3.6%	3.6% (earnings)	3.6% (earnings path)	£359.85	£358.45	0.86	0.62

On this path the figure that sets the triple lock changes four times from April 2030, and the rules differ in two years: April 2033 (triple lock 4.0%, earnings; plan 3.6%, rising to its earnings path) and April 2034 (triple lock 9.9%, earnings; plan 9.8%, rising to its earnings path). In April 2032 CPI runs ahead of earnings and both rules pay it; that lifts the plan above its earnings path, and the next year, when earnings lead, the plan rises only as far as its earnings path. In April 2034 the plan's rate is 0.1 point below the triple lock's although both follow earnings: the year before, the plan's rise to its earnings path was rounded up, leaving it just above the path, so it needed slightly less than the earnings rise. By 2039-40 the full new State Pension is £411.59 a week under the triple lock and £409.63 under the plan.

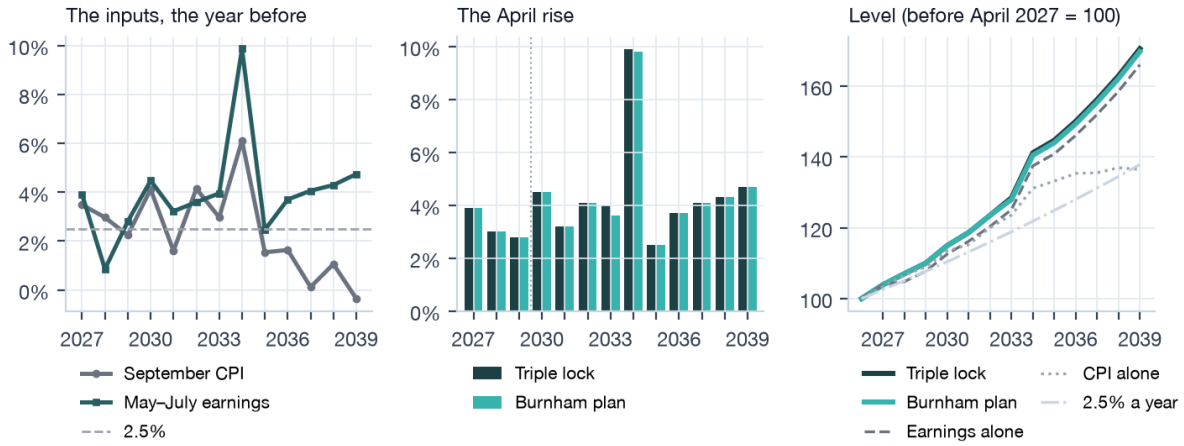


Figure 3: The two rules on a path drawn at random from the monthly model, shifted to the OBR's means. Panels as in the central-path figure. Rule arithmetic.

4.2.2 The middle and 90th-percentile paths

For context, Figure 4 adds two more paths, each picked as the one nearest the average of the paths close to a given point in the distribution of the 2039-40 gap: the middle (a gap of £16.67 a week) and the 90th percentile (£29.84). They are scenarios, not forecasts: each is one path the model makes. On the middle path the plan saves £10.31bn in 2039-40 (£7.42bn net), and on the 90th-percentile path £18.46bn (£12.88bn net). Table 6 sets out all four paths.

Table 6: Every path shown, in the final year: the full new State Pension a week (rule arithmetic), the saving (full model runs on the Enhanced FRS, £ billion), and the share of households with more than £1 a year less income than under the triple lock and their mean loss.

Path	Weekly, triple lock	Weekly, plan	Gap	Gross	Net	Households worse off	Mean fall in income, £ a year
Central forecast	£359.85	£358.45	£1.40	0.86	0.62	27.0%	£69
A random path	£411.59	£409.63	£1.96	1.21	0.86	27.0%	£94
Monthly model: middle	£361.13	£344.45	£16.67	10.31	7.42	27.1%	£822
Monthly model: 90th percentile	£392.37	£362.53	£29.84	18.46	12.88	27.1%	£1,437

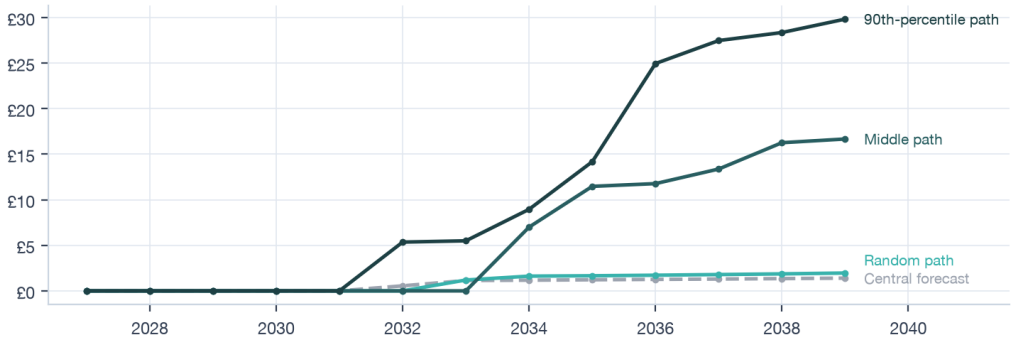


Figure 4: The gap between the two rules' full new State Pension on each path shown, £ a week. The central and random paths lie close together near zero. Rule arithmetic.

4.3 The expected saving

A single path cannot give the saving to expect. The saving comes from years in which CPI or the floor runs ahead of earnings, and the OBR's central path has almost none after the switch (Section 4.1). So we average the saving over the model's paths of prices and earnings (Section 4.2), estimated from full runs of both rules on 160 of the 50,000 paths. The result is conditional on that one model of prices and earnings: it is not a forecast probability, and it does not cover the choice of model. We report it because the model passes the test we registered before scoring it, in a dry run on today's data (Section 3.6); the binding test runs on the 28 October 2026 Budget forecast, and if the model fails it, the update removes the expected saving and says why.

Figure 5 and Table 8 give the expected saving by year with its Monte Carlo standard error. That error measures how precisely the runs pin down the average over all the paths, not the range of outcomes. In 2039-40 the expected saving is £11.14bn (SE £0.07bn) gross and £7.86bn (SE £0.05bn) net. Of the gross saving, extra Pension Credit offsets £0.61bn, lower income tax £2.46bn and extra Housing Benefit £0.15bn. In 2034-35, five rises after the switch, the expected gross saving is £3.21bn (SE £0.22bn). The middle 80% of paths runs from £3.9bn to £18.9bn gross in 2039-40.

By 2039-40 the full new State Pension is £931 a year (£17.91 a week) lower under the plan on average across the paths. The gap runs from £329 a year at the 10th percentile of paths to £1,583 at the 90th (rule arithmetic). Figure 6 shows each full run's 2039-40 saving against its weekly gap. The gross saving is £0.619bn for each £1 a week of gap in every run, because it comes only from the flat rates. The net saving scatters around a lower line, because the benefits that respond have thresholds.

4.3.1 Single scenarios

Beside the average we show a few named scenarios, each a full run of both rules (Table 7). The OBR's central forecast and the OBR's long-term path, on which the triple lock rises 0.6 points a year faster than earnings (Section 5.3), are single runs with no Monte Carlo error. The two earnings scenarios use the same simulated paths as the expected saving, with calendar-year earnings growth moved 0.5 percentage points a year below or above from 2031, and are averages with a Monte Carlo error. Without the shift those paths average £11.1bn; lower earnings growth

changes the 2039-40 saving by −£0.7bn and higher by −£0.1bn. The higher-earnings change is within its Monte Carlo error, so the two scenarios are not mirror images. The scenarios are a range of cases, not a probability interval.

Table 7: Single scenarios: the saving from the Burnham plan against the triple lock, £ billion, gross and net of tax and benefits. Single runs have no Monte Carlo error; path averages show ±1.96 standard errors.

Scenario	Run	2034-35 gross	2034-35 net	2039-40 gross	2039-40 net
OBR central forecast	single run	0.7	0.5	0.9	0.6
OBR long-term path (triple lock above earnings)	single run	3.2	2.3	10.3	7.3
Earnings −0.5 points from 2031	path average	2.8 ± 0.5	2.0 ± 0.3	10.4 ± 0.5	7.4 ± 0.4
Earnings +0.5 points from 2031	path average	3.5 ± 0.4	2.5 ± 0.3	11.0 ± 0.5	7.7 ± 0.4

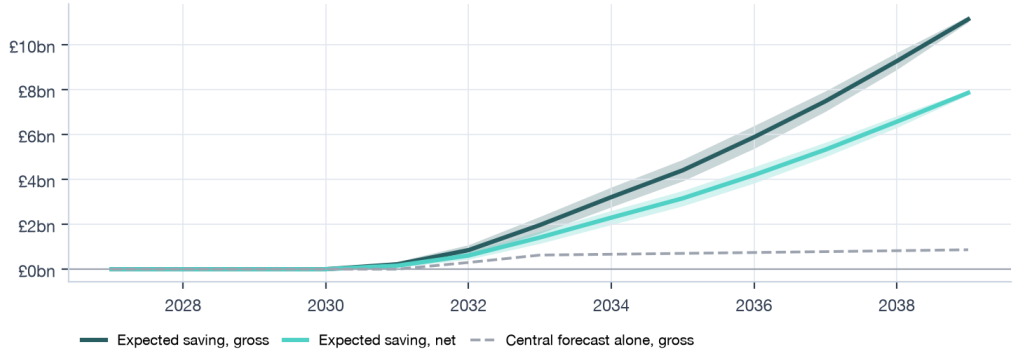


Figure 5: The expected saving from the Burnham plan by fiscal year, gross (State Pension spending) and net (the government's balance), with ±1.96 Monte Carlo standard errors (shaded: the precision of the average, not the range of outcomes), and the gross saving on the central forecast alone. Weighted means of full model runs on the Enhanced FRS.

4.4 Fiscal and distributional effects

The same random path runs through PolicyEngine UK for the whole survey population, once under each rule. Every pensioner's State Pension is recalculated, and with it their income tax, Pension Credit, Housing Benefit, council tax reduction and Winter Fuel Payment. The gross saving is the fall in spending on the basic and new State Pension; the net saving is the change in the government's balance once taxes and benefits respond. Table 6 gives the same measures on the middle and 90th-percentile paths.

4.4.1 Gross and net saving by year

On the random path the plan first saves money in 2033-34, £0.66bn gross, and by 2039-40 it saves £1.21bn gross and £0.86bn net (Figure 7, Table 9).

Table 8: The expected saving by fiscal year, £ billion, with Monte Carlo standard errors (SE), the expected share of households with more than £1 a year less income than under the triple lock, and the gross saving on the central path alone.

Year	Gross	SE	Net	SE	Households worse off	Central path, gross
2030-31	0.00	0.00	0.00	0.00	0.0%	0.00
2031-32	0.22	0.06	0.16	0.04	3.9%	0.00
2032-33	0.85	0.12	0.61	0.09	12.3%	0.30
2033-34	1.97	0.19	1.41	0.14	19.0%	0.63
2034-35	3.21	0.22	2.30	0.16	22.3%	0.66
2035-36	4.41	0.24	3.16	0.17	23.8%	0.70
2036-37	5.89	0.26	4.20	0.18	25.0%	0.74
2037-38	7.48	0.23	5.33	0.16	26.4%	0.78
2038-39	9.27	0.19	6.57	0.13	26.7%	0.82
2039-40	11.14	0.07	7.86	0.05	27.0%	0.86



Figure 6: Every full run behind the expected saving: the saving in the final year against the gap in the full new State Pension on that path. One point per run; the runs carry different weights in the estimate (by stratum).

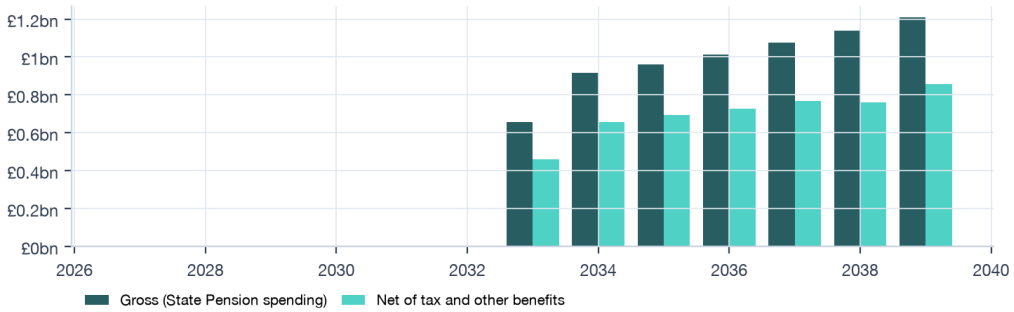


Figure 7: Gross and net saving each fiscal year on the random path, £ billion. Full model runs on the Enhanced FRS.

Table 9: The random path by fiscal year, from the switch: savings (£ billion) and households with more than £1 a year less income than under the triple lock, full model runs on the Enhanced FRS.

Year	Gross	Net	Households worse off	Mean fall in income, £ a year
2030-31	0.000	0.000	0.0%	£0
2031-32	0.000	0.000	0.0%	£0
2032-33	0.000	0.000	0.0%	£0
2033-34	0.658	0.461	25.6%	£58
2034-35	0.916	0.658	25.6%	£78
2035-36	0.961	0.695	26.1%	£81
2036-37	1.015	0.728	26.3%	£83
2037-38	1.075	0.767	26.8%	£86
2038-39	1.140	0.759	26.9%	£90
2039-40	1.210	0.856	27.0%	£94

4.4.2 From gross to net

Table 10 follows the gross saving to the net saving. In 2039-40 extra Pension Credit takes £0.07bn of the £1.21bn gross saving back, lower income tax £0.27bn and the other offsets £0.01bn, leaving £0.86bn: 71% of the gross.

Several features of the runs move the offsets, in both directions. In the survey runs Housing Benefit and council tax reduction respond only for households already receiving them. Nobody the plan makes newly entitled starts claiming, which makes their offsets smaller and the net saving larger. policyengine-uk lets a pension-age family make a new Housing Benefit claim, but it takes up either benefit only for a family that reports it, because the survey always has some family reporting a benefit. Pension Credit take-up is the dataset's own draw. A lower pension can make a household entitled to the guarantee credit. In the model it then gets the Housing Benefit passport only if it actually receives the guarantee credit, as in law (policyengine-uk #1927, in release 2.123.6). Earlier releases gave the passport on entitlement alone. Rents and council tax stay at their 2030 amounts after 2030, which lowers the most either benefit can pay and probably shrinks both offsets too. The survey has more Pension Credit and pension-age Housing Benefit than DWP forecasts (Section 5.1), which leans the other way: it makes the offsets larger and the net saving smaller.

4.4.3 Who has lower income

In 2039-40, 27.0% of households have a lower income under the plan on this path (by more than £1 a year), by £94 a year on average. On the middle path the share is 27.1% and the mean loss is £822. By income quintile (Table 11) the mean loss is largest in the 2nd and 3rd quintiles and smallest in the 5th; relative to income the 1st quintile sees the largest fall in income, 0.069% of its income.

Table 12 gives poverty after housing costs. The absolute line is fixed in real terms. The relative line is 60% of median household income in the same run, so when pensions fall the line can fall too. PolicyEngine weights that median by household, where the official statistics weight it by person.

Table 10: From gross to net on the random path, £ billion. Each offset is signed so the column adds up to the net saving.

	2034-35	2039-40
Saving on the basic and new State Pension (gross)	0.916	1.210
Extra Pension Credit	−0.063	−0.072
Extra Housing Benefit	−0.003	−0.003
Extra council tax reduction	0.000	−0.004
Extra Universal Credit	0.000	0.000
Extra Winter Fuel Payment	−0.001	−0.002
Less income tax	−0.191	−0.273
Everything else (other taxes and benefits)	0.000	−0.001
Saving to the government (net)	**0.658**	**0.856**

Table 11: Change in household income by income quintile on the random path in the final year. Quintiles are pairs of PolicyEngine's deciles of equivalised household net income under the triple lock; the deciles each hold a tenth of people, so quintiles hold unequal shares of households.

Quintile	Mean change, £ a year	Change, % of income	Total, £ billion	Share of households
Quintile 1	−£20.01	−0.069%	−0.153	22.7%
Quintile 2	−£34.89	−0.069%	−0.243	20.7%
Quintile 3	−£29.14	−0.042%	−0.186	19.0%
Quintile 4	−£25.72	−0.027%	−0.156	18.0%
Quintile 5	−£17.80	−0.012%	−0.117	19.6%

Table 12: Poverty after housing costs on the random path in the final year: share of people in households below the line. The change is the difference of the rounded figures.

Measure	Triple lock (current policy)	Burnham plan	Change, points
Pensioners, absolute	6.27%	6.31%	0.04
Pensioners, relative	10.95%	11.01%	0.06
Everyone, absolute	15.80%	15.81%	0.01
Everyone, relative	20.56%	20.58%	0.02

4.5 Example households

The same random path runs four example pensioners through PolicyEngine UK as households, once under each rule (Figure 8, Table 13). Each receives the full flat-rate State Pension, claims everything it is entitled to, and is the stated age in every year. The examples describe a pensioner of that age in each year, not one person ageing. Their private pensions, rents and council tax are set in 2026-27 terms and grow with the path's CPI. The renters make a new Housing Benefit claim, as the law allows once every adult in the family is over State Pension age. They report no existing claim, and policyengine-uk lets a pension-age family make a new claim. The examples are hypothetical households, not survey records. On the middle path the difference in State Pension in 2039-40 is £867 a year, against £102 on this one.

A lower State Pension means less income tax for a pensioner above the personal allowance. It means more Pension Credit for one entitled to it, and more council tax reduction or Housing Benefit for one receiving them. It can also bring a pensioner back under the Winter Fuel Payment's income threshold. In 2039-40 a pensioner on the new State Pension receives £101.72 a year less State Pension under the plan on this path.

The pensioner with a private pension (pensioner (b), single pensioner aged 70 with a £15,000 private pension, owns their home) pays £20.34 less income tax and has the rest as lower income, 80% of the loss.

The pensioner with the State Pension only (pensioner (a), single pensioner aged 70, State Pension only, owns their home) has an income after tax below the Pension Credit guarantee in 2039-40. So Pension Credit rises by £81.37. That is exactly the fall in the pension after the £20.34 fall in income tax. Pension Credit counts income after tax, both in the regulations (State Pension Credit Regulations, 2002, regulation 17(10)) and in policyengine-uk. The two together therefore make up the whole difference, and the pensioner comes out even. The council renter on the same pension (pensioner (c)) comes out the same way. In 2033-34, 2034-35 and 2035-36 their income after tax is above the guarantee or only just below it, so both have lower income for part of the difference (Figure 8).

Council tax reduction no longer moves for these two pensioners. The regulations disregard the whole income of a person receiving the guarantee credit (Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations, 2012, Schedule 1, paragraph 13). policyengine-uk has done the same since release 2.104.2, so their council tax reduction is the same under both rules.

The pensioner on the old basic State Pension (pensioner (d), single pensioner aged 90 on the old basic State Pension, rents from the council at £120 a week) has a smaller difference in State Pension, £77.94. Pension Credit and income tax replace all of it in 2039-40. In 2037-38, 2038-39 and 2039-40 this pensioner does have lower income for part of the difference. Its Pension Credit then includes savings credit, which pays 60% of income above a threshold. A lower pension therefore also lowers the savings credit, and Pension Credit rises by only 40% of the after-tax loss.

The renters' Housing Benefit does not change in any year on this path: it already covers the full rent, so it cannot rise.

The examples: (a) single pensioner aged 70, State Pension only, owns their home; (b) single pensioner aged 70 with a £15,000 private pension, owns their home; (c) single pensioner aged 70, State Pension only, rents from the council at £120 a week; (d) single pensioner aged 90 on the old basic State Pension, rents from the council at £120 a week.

In policyengine-uk the Winter Fuel Payment is paid, outside Scotland, to pensioners on

Table 13: Example pensioners on the random path in the final year: the change under the Burnham plan, £ a year (income tax: the change in tax paid, so a fall adds to income). The last row is the change in income after tax and benefits as a share of the change in State Pension.

	(a)	(b)	(c)	(d)
State Pension	−£101.72	−£101.72	−£101.72	−£77.94
Income tax paid	−£20.34	−£20.34	−£20.34	−£15.59
Pension Credit	£81.37	£0.00	£81.37	£24.94
Housing Benefit	£0.00	£0.00	£0.00	£0.00
Council tax reduction	£0.00	£0.00	£0.00	£0.00
Winter Fuel Payment	£0.00	£0.00	£0.00	£0.00
Income after tax and benefits	£0.00	−£81.37	£0.00	−£37.41
Share of the lower pension felt in income	none	80%	none	48%

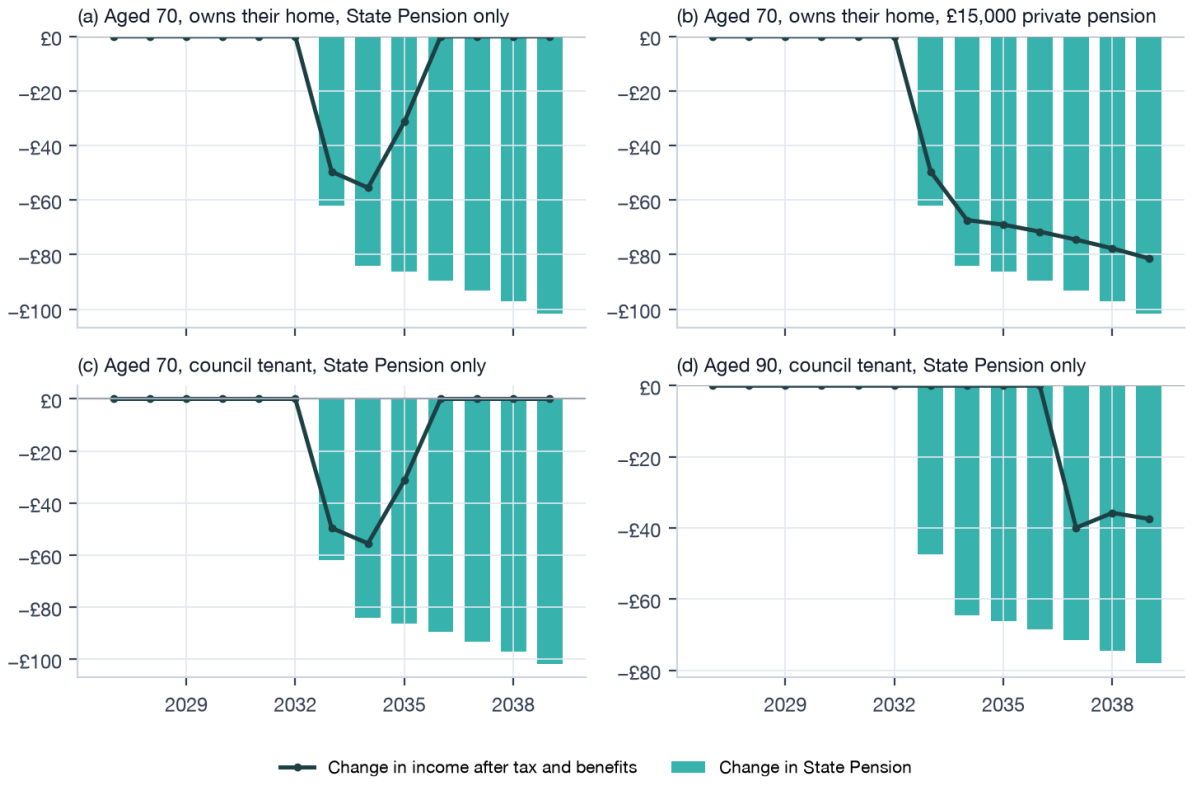


Figure 8: The example pensioners on the random path: the change in State Pension under the Burnham plan (bars) and in income after tax and benefits (line), £ a year, each fiscal year. Full PolicyEngine UK runs of each household under each rule.

Pension Credit or certain other income-related benefits and, in England and Wales, to households in which a pensioner's taxable income is below a fixed threshold. It changes for pensioner (b) in 2036-37 on the middle and 90th-percentile paths, where the plan's lower pension brings that pensioner's taxable income under the threshold and the payment of £200 offsets part of the difference.

On the 90th-percentile path the council renter shows a larger step. In 2039-40 its State Pension is £1,552 lower under the plan, which brings it onto the Pension Credit guarantee credit. A pensioner receiving the guarantee credit has no income counted in the Housing Benefit means test, so its Housing Benefit rises by £1,504 to the maximum. Its income rises by £1,390. The step is this large partly because of how the runs uprate. They raise the Pension Credit guarantee with earnings, as the law requires at least, while policyengine-uk raises the Housing Benefit personal allowance for pensioners with CPI. The allowance starts above the guarantee. On paths where earnings outgrow prices the guarantee overtakes it. From then on a pensioner moving onto guarantee credit moves from a tapered Housing Benefit award to the maximum in one step. This example claims Pension Credit, so it receives the guarantee credit and the step is real in law. In the survey runs the step now happens only for a household that receives the guarantee credit, not one that is merely entitled to it (Section 3).

5 Discussion

5.1 The survey data against DWP

The gross saving scales with State Pension spending in the data. Every run checks that each flat-rate pension moves exactly in proportion to the flat rate. So the closer the data's State Pension bill is to DWP's, the more the gross saving can be trusted. Table 14 sets the Enhanced FRS against DWP's Spring Forecast 2026 (Department for Work and Pensions, 2026a) for 2026-27. DWP's tables cover Great Britain and include pensions paid abroad. We take the paid-abroad amounts out of DWP's totals where the tables separate them, and count only households in England, Scotland and Wales in the model, so the two match. No DWP figure exists after 2030-31.

The survey's State Pension spending is 15% below DWP's, so the gross saving, which scales with it, is if anything on the low side. Pension Credit is 17% above DWP's and pension-age Housing Benefit 3% above, so the benefit offsets are if anything on the high side, and the net saving correspondingly a little low.

5.2 DWP's costing

DWP costs the plan at £15bn in 2039-40 in cash terms (£11bn in 2025-26 prices) (Department for Work and Pensions, 2026b). Our expected gross saving is £11.1bn for the UK. Our figures differ from DWP's for four reasons.

The first is the path. DWP applies one uprating path, which it does not state. Ours are a set of paths calibrated to the OBR's forecast. On the central path alone we find £0.86bn. For a single path to save £15bn it needs several years in which CPI or the floor beats earnings. Our 90th-percentile path saves £18.46bn gross.

The second is the population. DWP's model, Pensim3, projects the population and each person's contribution record. We reweight the survey instead (Section 3.3). Each survey record stands for a person of its age in every year. The weights follow the ONS projection of the

Table 14: The survey data against DWP's Spring Forecast 2026, Great Britain, matched where DWP's tables separate payments abroad.

2026-27	DWP forecast	Enhanced FRS, Great Britain
State Pension spending (in GB, excluding payments abroad), £bn	£148.3bn	£125.9bn
Basic and new State Pension, £bn (DWP's figure includes payments abroad)	£131.0bn	£112.1bn
State Pension recipients (in GB), millions	12.15m	11.16m
Pension Credit, £bn	£6.1bn	£7.1bn
Pension Credit claims (benefit units), millions	1.31m	1.48m
Housing Benefit (not Universal Credit housing), £bn	£12.5bn	£8.2bn
Housing Benefit, pension age, £bn	£7.3bn	£7.5bn
Housing Benefit claims, pension age (benefit units), millions	1.08m	1.11m

population by age and sex, and each person's State Pension type follows from their date of birth. So later pensioners join on the new State Pension. But their amounts come from today's records of the same age, and deaths, migration and contribution histories are not simulated.

The third is coverage. The survey's State Pension spending is 15% below DWP's matched figure (Table 14), and the gross saving scales with it.

The fourth is scope. DWP's figure is direct spending for Great Britain, with no tax effects, rounded to the nearest £1bn. It compares with our gross figure for Great Britain, which the results give beside the UK one. DWP also cautions that "These savings should not be used to then derive further point estimates for savings in an individual year and should instead be relied on for long-term estimates" (Department for Work and Pensions, 2026b).

5.3 The OBR's long-term premium

The OBR's long-term projections assume the triple lock raises the pension 0.6 percentage points a year faster than earnings (Office for Budget Responsibility, 2026d). On our paths the triple lock rises 0.55 points a year faster than May–July earnings on average over the April rises of 2034 to 2039; the Burnham plan rises 0.08 points a year slower. Across the calibrations that match the OBR's means the triple lock's premium runs from 0.43 to 0.60 points. This is rule arithmetic on the inputs, not a fiscal figure, and it compares the triple lock's rise with the statutory earnings measure, where the OBR's premium is over its own measure of average earnings.

5.4 Other published costings

Table 15 sets our closest figures beside other published costings. Only DWP's costing is of this plan; the others compare the triple lock with earnings uprating, over the years since 2012 or over a long horizon, so they are context, not checks. The OBR estimated in July 2025 that the triple lock would cost £15.5 billion a year by 2029-30 against earnings uprating since 2012 (Office for Budget Responsibility, 2025). The Resolution Foundation cites the OBR's estimate that State Pension spending would be £12.6bn lower this year with earnings-linked uprating than with the triple lock, and puts the net saving at about £9bn after income tax and means-tested benefits (Curtice and Clegg, 2026). Had the Burnham plan, which keeps a floor of the higher of CPI and 2.5%, started in April 2012, 2013, 2014 or 2015, we find it would save £7.25bn gross

and £5.24bn net in 2026-27. The OBR's July 2026 report looks further ahead: state pension spending is "projected to rise from 5 per cent of GDP to around 9 per cent of GDP over the projection period in the baseline", driven by ageing and triple-lock uprating, and with earnings uprating instead it "reaches around 7 per cent of GDP" (Office for Budget Responsibility, 2026b). The closest methodological precedent is the Institute for Fiscal Studies' simulation of the triple lock's cost from historical inflation and earnings (Cribb et al., 2023).

Table 15: Other published figures and our closest figure. 'Like for like' says whether the two measure the same thing.

Source	Their figure	Our closest figure	Like for like
Department for Work and Pensions, *State Pension uprating analysis 2026* (2026-09-29)	Savings against the triple lock of £15bn in 2039-40 in cash terms, £11bn in 2025-26 prices	£11.14bn	Partly
Office for Budget Responsibility, *Long-term economic determinants, March 2026 EFO* (2026-05-28)	The 'Triple lock' growth rate: "Average earnings growth plus 0.6 percentage points"	0.55 points a year	Partly
Office for Budget Responsibility, *Fiscal risks and sustainability - July 2025* (2025-07-08)	"…will have added £15.5 billion (0.5 per cent of GDP) to state pension spending annually by 2029-30", against "the £5.2 billion we estimate the triple lock would have cost by that point under initial assumptions"	£7.25bn	No
Resolution Foundation, *What a ratchet! (Curtice and Clegg)* (2026-06-10)	State Pension spending £12.6bn lower this year with earnings-linked uprating (the OBR's estimate); about £9bn net	£5.24bn	No
Office for Budget Responsibility, *Fiscal risks and sustainability - July 2026* (2026-07-07)	State pension spending "projected to rise from 5 per cent of GDP to around 9 per cent of GDP over the projection period in the baseline"; with earnings uprating it "reaches around 7 per cent of GDP"	£11.14bn	No

AME: Annually Managed Expenditure.

5.5 Limitations

Our results depend first on how the plan is defined. The speech gave no formula. We read DWP's description as a rule that, from April 2030, raises the pension by at least the higher of CPI and 2.5%, and by whatever more keeps it at its 2029-30 ratio to earnings, with the top-up rounded up to 0.1 point. Every rule follows the triple lock to April 2029, and each rise uses the triple lock's measures, September CPI and May–July earnings, although the statute requires only the earnings link. Only the basic and new State Pension differ between the rules. Under both, the additional State Pension rises with September CPI, the Pension Credit guarantee with May–July earnings (its statutory minimum), and State Pension age follows the law's timetable by date of birth. Costs are in cash terms, for the UK and for Great Britain, and we model no behavioural response.

Second, any statement about uncertainty rests on one statistical model of prices and earnings, a monthly VAR fitted to 2000–2026 and shifted so that its calendar-year averages equal the OBR's. No form of the model passed the screen we first set, because of the 2021 furlough base effect. We changed the screen after seeing those scores, to score April 2022 as the law applied it. We registered the new rule before scoring it, and we report the first failure beside it, but the change was still made after the fact. The model passes the new screen in a dry run on today's data; the binding score uses the 28 October 2026 Budget forecast. The expected saving is conditional on that model and is not a forecast probability, and the single scenarios are named cases, not a range for the saving. The backtests rest on six and twelve overlapping four-year windows, using today's revised data rather than the figures available at the time. Its statutory inputs depart from the OBR's quarterly figures before the switch. After 2030 the central path follows the OBR's long-term path, which the OBR describes as not a forecast. In every run April 2027's benefit uprating uses the model's calendar-2026 CPI forecast, not the September 2026 figure.

Third, the population is aged statically, not projected. Each record stands for a person of its age in every year. Ages above the survey's top code of 80 are assigned, not observed. The weights follow the ONS projection by age and sex, and pension types follow date of birth, but later pensioners' amounts come from today's records of the same age. Deaths, migration and contribution histories are not simulated, and pensioners living abroad are outside the survey. A record that moves from the basic to the new State Pension keeps its reported amount by default. Its legal amount cannot be rebuilt from the survey, so the full new State Pension is shown only as a bound on the flat-rate part. The data also carry a known weight gap (policyengine-uk-data #538): the data build calibrates its weights to 2025, but policyengine-uk uprates them back to 2025 by its own population growth, so every 2025 total is 0.44% low while the relative weights are exact. Our ageing keeps the model's own 2025 weights and grows them from there, so it carries the gap into every later year, neither adding to it nor closing it. The fix belongs to the data build or the model. Rents and council tax are held at their 2030 amounts after 2030, and dividend, property, savings and self-employment income do not follow the simulated path.

Fourth, we can check the data against DWP only in part. DWP's tables match our Great Britain figures only to 2030-31, where they stop. They give no matched figure for the basic and new State Pension separately, and no forecast by age or by country. Recipient counts, the split between basic and new State Pension, and forecasts are not published by country, so a country check cannot be done from published data.

Finally, the net figures, though not the gross ones, depend on how policyengine-uk models the benefits that respond to a lower pension. In the survey runs Housing Benefit and council tax reduction respond only for households already receiving them, so nobody the plan makes newly entitled starts claiming. This understates the offsets and overstates the net saving. The Housing Benefit passport for guarantee credit recipients now keys on receipt, as in law. But the runs raise the Pension Credit guarantee with earnings while policyengine-uk raises the pensioner Housing Benefit personal allowance with CPI. On paths where earnings outgrow prices the guarantee overtakes the allowance. A pensioner who then starts receiving the guarantee credit gains the maximum Housing Benefit in one step, so a small pension cut can move one heavily weighted survey household's income by a large amount. policyengine-uk does not model the basic State Pension's age addition at 80 (policyengine-uk #2139), and it pays Scotland's Pension Age Winter Heating Payment at a flat household rate rather than the rates in the regulations (policyengine-uk #2138).

6 Conclusion

The Burnham plan keeps the triple lock's protection against prices and its 2.5% floor, but not its carry-forward of higher rises. After a year in which the higher of CPI and 2.5% outpaces earnings, the triple lock keeps the extra rise for good, while the plan waits for earnings to catch up. Its saving therefore depends on how often the lead passes between prices and earnings, not only on their expected growth. On the OBR's central forecast the plan saves £0.9bn in State Pension spending in 2039-40. Averaged over the model's paths the expected saving is £11.1bn gross and £7.9bn net, provisional until the binding test on the Budget forecast.

About 27% of households have a lower income under the plan in 2039-40. Pension Credit and income tax offset part of the difference. Pensioners on the Pension Credit guarantee come out even, and those with private pensions have lower income for most of it: a pensioner with a £15,000 private pension keeps 80% of the difference as lower income.

The general lesson is for costing any uprating rule that takes the maximum of several indices. Its cost is a property of the distribution of the indices, and a single central path, on which the indices rarely cross, can understate it. Stating that cost as an expected value needs a model of the indices that passes a test set in advance, and we report whether ours did. Our survey is aged statically, by reweighting to the ONS projection and assigning pension types by date of birth, but it does not simulate deaths, migration or contribution histories. A costing that combined a dynamic population model, such as DWP's, with a distribution of paths rather than one would narrow that gap.

References

BBC News (2026). How much could Burnham's change to triple lock save? Entry on the BBC News live page, published 29 September 2026 at 17:52 BST. A live page, updated after publication; the entry was still on it when read on 30 September 2026.

Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations (2012). Council Tax Reduction Schemes (Prescribed Requirements) (England) Regulations. SI 2012/2885, Schedule 1, Part 6, Chapter 2, paragraph 13. Accessed 30 September 2026.

Cribb, J., Emmerson, C., and Karjalainen, H. (2023). The triple lock: uncertainty for pension incomes and the public finances. IFS Report R272, Institute for Fiscal Studies. Accessed 30 September 2026.

Curtice, R. and Clegg, A. (2026). What a ratchet! Why it's time to stop being polite about the triple lock. Technical report, Resolution Foundation. Published 10 June 2026. Accessed 30 September 2026.

Department for Work and Pensions (2026a). Benefit expenditure and caseload tables 2026. Outturn and forecast tables: Spring Forecast 2026. Published 14 April 2026. Accessed 30 September 2026.

Department for Work and Pensions (2026b). State Pension uprating. Research and analysis, in collection "state pension uprating analysis 2026", Department for Work and Pensions. Published 29 September 2026. Accessed 30 September 2026.

Gneiting, T. and Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477):359–378.

Neyman, J. (1934). On the Two Different Aspects of the Representative Method: The Method of Stratified Sampling and the Method of Purposive Selection. *Journal of the Royal Statistical Society*, 97(4):558–606.

Office for Budget Responsibility (2025). Fiscal risks and sustainability – July 2025. Technical report, Office for Budget Responsibility. Published 8 July 2025. Accessed 30 September 2026.

Office for Budget Responsibility (2026a). Economic and fiscal outlook – March 2026. Technical report, Office for Budget Responsibility. Published 3 March 2026. Detailed forecast tables, economy: https://obr.uk/docs/d055fbf02d5b3g6jq8l2/efo-march-2026-detailed-forecast-tables-economy.xlsx. Accessed 30 September 2026.

Office for Budget Responsibility (2026b). Fiscal risks and sustainability – July 2026. Technical report, Office for Budget Responsibility. Published 7 July 2026. Accessed 30 September 2026.

Office for Budget Responsibility (2026c). Historical official forecasts database – March 2026. Spreadsheet (Historical_official_forecasts_database_Spring_2026.xlsx). Listed at https://obr.uk/data/, dated 26 March 2026. Accessed 30 September 2026.

Office for Budget Responsibility (2026d). Long-term economic determinants – March 2026 Economic and fiscal outlook. Spreadsheet, dated 28 May 2026 (sheets "Long-term economic determinants" and "Growth rate assumptions"). Accessed 30 September 2026.

Office for National Statistics (2026a). AWE: Whole Economy Level (£): Seasonally Adjusted Total Pay Excluding Arrears. Time series KAB9, dataset LMS, release of 15 September 2026. Accessed 30 September 2026.

Office for National Statistics (2026b). AWE: Whole Economy Year on Year Three Month Average Growth (%): Seasonally Adjusted Total Pay Excluding Arrears. Time series KAC3, dataset LMS, release of 15 September 2026. Accessed 30 September 2026.

Office for National Statistics (2026c). CPI ANNUAL RATE 00: ALL ITEMS 2015=100. Time series D7G7, dataset MM23, release of 16 September 2026. Accessed 30 September 2026.

Office for National Statistics (2026d). CPI INDEX 00: ALL ITEMS 2015=100. Time series D7BT, dataset MM23, release of 16 September 2026. Accessed 30 September 2026.

Office for National Statistics (2026e). National population projections: 2024-based, UK, principal projection. Released 28 April 2026.

Pensions Act (2014). Pensions Act. 2014 c. 19, section 26. Accessed 30 September 2026.

PolicyEngine (2026a). policyengine. Python package, version 5.3.0. Source: https://github.com/PolicyEngine/policyengine.py. Accessed 30 September 2026.

PolicyEngine (2026b). policyengine-uk. Python package, version 2.90.2. Source: https://github.com/PolicyEngine/policyengine-uk. Accessed 30 September 2026.

Robertson, J. C., Tallman, E. W., and Whiteman, C. H. (2005). Forecasting Using Relative Entropy. *Journal of Money, Credit and Banking*, 37(3):383–401.

Scheuerer, M. and Hamill, T. M. (2015). Variogram-Based Proper Scoring Rules for Probabilistic Forecasts of Multivariate Quantities. *Monthly Weather Review*, 143(4):1321–1334.

Social Security Administration Act (1992). Social Security Administration Act. 1992 c. 5, section 150A. Accessed 30 September 2026.

Social Security Benefits Up-rating Order (2011). Social Security Benefits Up-rating Order. SI 2011/821, article 5(3): the basic pension in a Category A retirement pension rises from £97.65 to £102.15 a week. Accessed 30 September 2026.

Social Security (Up-rating of Benefits) Act (2021). Social Security (Up-rating of Benefits) Act. 2021 c. 32, section 1: for the tax year ending with 5 April 2022 the section 150A review is "by reference to the general level of prices" instead of earnings. Accessed 30 September 2026.

State Pension Credit Regulations (2002). State Pension Credit Regulations. SI 2002/1792, regulation 17(10): "there shall be disregarded– (a) any amount payable by way of tax". Accessed 30 September 2026.

A The tests on past forecasts

This appendix gives the scores behind the screen of Section 3.6, then an earlier, wider backtest kept as the record.

Table 16 gives the dry-run scores of the screen on today's inputs, for each of the five forms in tests A and B. April 2022 is scored as the law applied it (earnings set equal to CPI), which decides the verdict, and as published, which is reported beside it and has no say. Coverage is the share of outcomes inside the form's central 80% band: of the yearly gaps between earnings and CPI ("annual") and of the gap the two rules open after four rises ("final years"). The biases are the form's mean minus what happened: in the yearly gap (percentage points), in the number of changes of lead per path, and in the share of years with earnings below the floor. A form must pass every condition in both tests; the verdict column gives the screen's result for the form as a whole.

Table 16: The screen's dry-run scores on today's inputs: coverage of the central 80% band and biases, by test, form and treatment of April 2022. The verdict uses April 2022 as in law; the published treatment is shown for reference only.

Test	Form	April 2022	Verdict	Annual coverage	Final years inside	Gap bias	Lead-change bias	Floor-share bias
A	Monthly model (used here)	As in law	Pass	80%	4 of 6	1.36	−0.17	0.24
A	Monthly model (used here)	As published	Reference	67%	1 of 6	0.47	−0.90	0.18
A	Monthly model, two lags	As in law	Pass	65%	4 of 6	1.36	−0.21	0.22
A	Monthly model, two lags	As published	Reference	54%	1 of 6	0.47	−0.95	0.15
A	Annual model plus past gaps	As in law	Fail	80%	4 of 6	1.36	−0.37	0.24
A	Annual model plus past gaps	As published	Reference	67%	1 of 6	0.47	−1.18	0.20
A	Monthly model, Student-t shocks	As in law	Pass	60%	4 of 6	1.36	−0.21	0.22
A	Monthly model, Student-t shocks	As published	Reference	50%	1 of 6	0.47	−0.96	0.15
A	Monthly model, Gaussian shocks	As in law	Pass	80%	4 of 6	1.36	−0.14	0.25
A	Monthly model, Gaussian shocks	As published	Reference	67%	1 of 6	0.47	−0.87	0.19
B	Monthly model (used here)	As in law	Pass	89%	10 of 12	1.35	−0.19	0.03
B	Monthly model (used here)	As published	Reference	81%	7 of 12	0.90	−0.56	0.00
B	Monthly model, two lags	As in law	Pass	73%	10 of 12	1.35	−0.26	0.00
B	Monthly model, two lags	As published	Reference	67%	7 of 12	0.90	−0.64	−0.03
B	Annual model plus past gaps	As in law	Fail	84%	10 of 12	1.36	−0.41	0.04
B	Annual model plus past gaps	As published	Reference	77%	7 of 12	0.91	−0.81	0.02
B	Monthly model, Student-t shocks	As in law	Pass	70%	10 of 12	1.35	−0.26	0.01
B	Monthly model, Student-t shocks	As published	Reference	65%	7 of 12	0.90	−0.63	−0.03
B	Monthly model, Gaussian shocks	As in law	Pass	89%	10 of 12	1.35	−0.17	0.04
B	Monthly model, Gaussian shocks	As published	Reference	81%	7 of 12	0.90	−0.53	0.01

Before that screen we ran a wider backtest, kept here as the earlier record. It is why the paper reports an expected value and scenarios, not the probability of any outcome. We scored nine ways of putting a range on the two statutory inputs, and the OBR forecast alone, against what followed six OBR forecasts (the March 2016 EFO to the March 2021 EFO): the forecasts with at least two earlier forecasts whose target years had all been published, which the method built on past forecast errors needs. Each method is fitted only on data dated before the forecast, using today's revised figures, and scored on the September CPI and May–July earnings for the four years after the forecast year, which set the April rises two to five years after it.

The scores are proper scoring rules (Gneiting and Raftery, 2007): the continuous ranked probability score (CRPS) for the gap between the triple lock and the Burnham plan after those rises, in points of the pension; the energy score for the two series over the years together; and the variogram score (Scheuerer and Hamill, 2015), which rewards getting the dependence between years right. Lower is better for each. We also record whether the realised gap fell inside each

method's middle 80%, and how many changes of lead between CPI and earnings the method expected against how many happened.

Table 17 gives each method's score for the gap and its coverage, with April 2022 scored both ways; how April 2022 is scored decides the result. Scored on the published May–July 2021 earnings growth of 8.4%, every method expected fewer changes of lead than happened, and the best placed one of six realised gaps in its middle 80%. Scored with earnings equal to CPI (3.1%) that April, as the law set it when the earnings leg was suspended, the best placed four of six; the best-scoring of those (monthly model, not calibrated) missed the March 2020 EFO and March 2021 EFO forecasts, whose rises include April 2023, set by September CPI of 10.1%. The method built on past OBR forecast errors has only 2–7 paths at these forecasts, so its middle 80% is the full range of those paths. Neither treatment gives a reliable 80% range, so we report at most an expected saving, and only from a model that passes the screen of Section 3.6. The spread across paths we describe as scenarios.

Table 17: The forecast-distribution backtest over the chronological OBR forecasts, with April 2022 scored as published and as in law: CRPS of the gap after the rises (points of the pension); how many realised gaps fell inside the method's middle 80%; the energy score and the variogram score over the same series in different years; and changes of lead, predicted against actual. Lower scores are better.

Method	Gap CRPS, published	Inside	Gap CRPS, in law	Inside	Energy, published	Variogram, published	Lead changes, published
Monthly, drift shift (used here)	1.88	1	1.07	4	7.87	11.40	1.3 / 2.2
Monthly, tilted	1.88	1	1.06	4	7.85	11.39	1.3 / 2.2
Monthly, not calibrated	1.97	1	1.04	4	7.76	11.46	1.3 / 2.2
Monthly, t-copula, tilted	2.09	1	1.15	4	8.09	12.18	1.2 / 2.2
Monthly, Gaussian, tilted	1.82	1	1.05	4	7.84	11.28	1.3 / 2.2
Annual, plus past gaps	2.13	1	1.15	4	7.56	10.52	1.0 / 2.2
Annual, calendar only	2.50	1	1.29	3	7.77	12.60	0.8 / 2.2
Past OBR errors	2.35	0	1.34	3	8.85	12.18	0.9 / 2.2
Independent normal	2.22	1	1.22	4	8.20	11.85	1.2 / 2.2
OBR forecast (a point)	3.20	n/a	1.63	n/a	10.04	29.88	0.0 / 2.2

B Additional results

B.1 If the plan had started earlier

Figure 9 applies the plan's rule to the published inputs from an earlier April. Started in April 2012, 2013, 2014 or 2015, it would have paid £226.11 a week in full new State Pension in 2026-27, against the £241.30 actually paid: 6.3% less. It pays less than the triple lock in April 2016, 2019, 2020, 2024, 2025 and 2026: each time the plan is still above its earnings path after earlier years in which CPI or the floor ran ahead of earnings. The largest shortfalls follow April 2023, when September CPI of 10.1% set a rise far above earnings growth of 5.5%. Started in April 2024, 2025 or 2026, the plan would have paid the same as the triple lock every year to April 2026. The plan's earnings path here also grows by CPI in April 2022, as if the suspension of the earnings link applied to it too; with the published figure its earnings path would have been higher from then on, and the shortfalls here smaller.

For 2024-25 to 2026-27, the years the survey data cover, the savings are full model runs in which every flat-rate pension is set to the plan's level instead of the triple lock's. Had the plan started in April 2012, 2013, 2014 or 2015, State Pension spending in 2026-27 would have been

£7.25bn lower, and the government would have been £5.24bn better off once income tax and benefits respond.

The same comparison tests the monthly model behind our paths, before any shift. Fitted only on data to December 2010, and with no calibration because no forecast covers those years, the monthly model of Section 4.2 gives a plan started in April 2012 an average gap of 6.1% of the pension by April 2026. What happened gives 6.3%, at the 56th percentile of the model's paths. This check is one part of the screen in Section 3.6, which scores it with April 2022 as the law applied it.

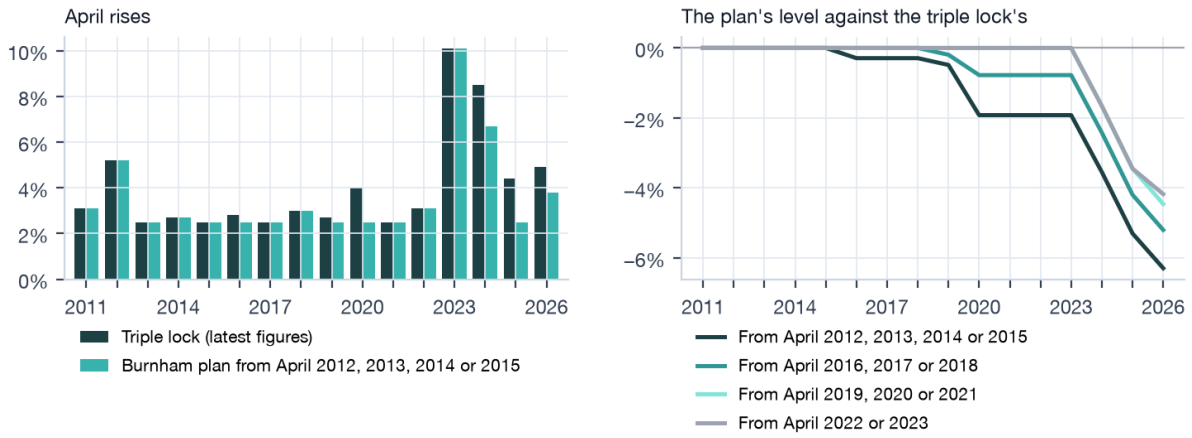


Figure 9: The Burnham plan's rule applied from earlier Aprils to the published inputs. Left: the April rises under the triple lock (latest figures) and under the plan started at the earliest date that changes anything. Right: the plan's level against the triple lock's, by start date. Rule arithmetic.

C Model details

C.1 Inputs held the same under both rules

Four inputs are set the same way under both rules, so that only the basic and new State Pension differ between the two runs of a path. The first is each person's State Pension type, from their date of birth as above. Each run reads the types back from the model every year and fails if any differs. The second is the additional State Pension, at its survey-year amount grown by September CPI (published to April 2026, then the path's), for people over State Pension age that year. policyengine-uk still scales the additional pension by the ratio of the flat rates (policyengine-uk #1941), so lower flat rates would lower it too. We hold it fixed instead. The third is State Pension age, the model's own as above. The fourth is the Pension Credit guarantee. It rises with May–July earnings and is never cut, the minimum the law requires (Social Security Administration Act, 1992). policyengine-uk uprates it by CPI. This is an assumption, the same under both rules.

C.2 Jobs and reproducibility

Each model run is a job in its own process. It is cached under a hash of its arguments, the analysis code and the package versions, so an interrupted build resumes and any change to the engine re-

runs every job. The results in this paper were built at revision 8838fd4 of PolicyEngine/uk-triple-lock, with the release bundle above. The manuscript reads `data/results.json` and computes every number it prints from it. The repository's README gives the commands to rebuild the file. The results file gives no survey record's identifier, weight or amount, and no single record's contribution to a total, because FRS records are licensed data.